

Building a Reproducible Scholarly Search Toolkit

Episode 0: The Architecture of Scientific Discovery

Nexus Scholar Suite

The Scientific Literature Crisis

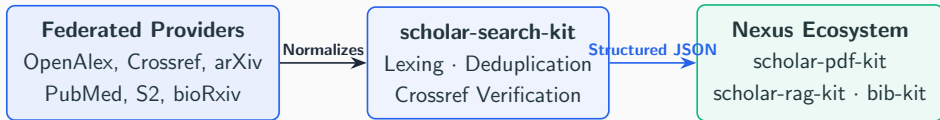
Proprietary Silos

- Black-box search ranking algorithms.
- Unreproducible result lists over time.
- Vendor lock-in and steep API costs.

Scientific Requirement

- Full query auditability & provenance.
- Multi-provider federated discovery.
- Deterministic deduplication trails.

Nexus Scholar Suite Architecture



Why Build from First Principles?

Auditability over Convenience

We cannot scientifically trust what we cannot trace. Building clean-room components ensures mathematical exactness for deduplication denominators and citation metrics.

- **Resilient Client:** Polite token buckets and SQLite session caching.
- **Anti-Hallucination:** Verification against 150M Crossref records.
- **Agentic Automation:** Native Antigravity skills for AI research assistants.

Let's build the scientific search backbone.